MOBILE COMMUNICATION CONCEPTUAL MAP

Understanding Mobile Network Protocol Stacks

Layers, Protocols and Communication Flow

A mobile network protocol stack consists of multiple layers that work together to transmit user data and control information between a mobile device, the radio access network, and the core network.

Mobile Station (MS)

Application Layer
Web apps, app browsing, messaging

Transport Layer
TCP / UDP

Network Layer
IP

Packet and Link Processing
RLC / PDCP / MAC

Physical Layer Explore
Radio signal transmission

Radio Access Network

Radio Resource Control
RRC connection setup

Packet Data Convergence Protocol
PDCP

Radio Link Control
RLC

Medium Access Control
MAC

Physical Layer Explore
Radio transmission

Core Network

IP Connectivity
Packet routing

Mobility Management
Tracking and mobility

Session Management
Data session handling

Authentication
Subscriber validation

External Services Explore
Internet and voice networks

Device: MS Mobile equipment and SIM

RAN: Radio eNodeB / gNodeB

Core: 5G Network services

ASSIGNMENT / PROJECT INFORMATION

About the Project

Visual Representation of Mobile Network Protocol Stacks

Project information

PROJECT	Mobile Network Protocol Stacks
STUDENT	Pranith
ORGANIZATION	SIMATS
SUBJECT	Mobile Computing
DESCRIPTION	This prototype presents the major protocol layers involved in mobile communication and explains how the mobile station, radio access network, and core network cooperate to deliver data and signaling.

Reading the presentation

Overview
Three network domains and their conceptual responsibilities.

Protocol Layers
Eight protocol summaries organized by layer and control plane role.

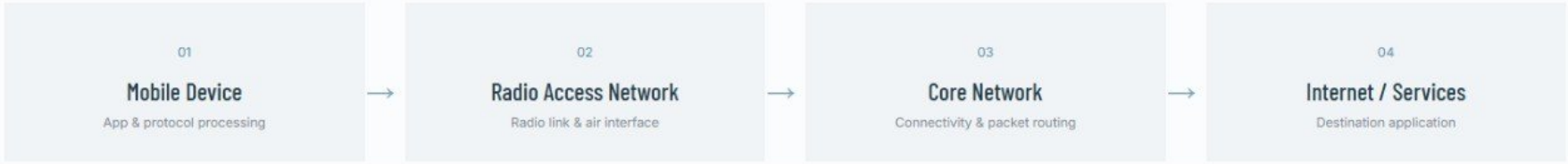
Communication Flow
A user-data request, its return response, and the signaling paths.

Academic note The diagrams describe conceptual layers and functions; cellular generations and architectures differ.

END-TO-END COMMUNICATION

How Mobile Communication Works

Follow the user-data journey and distinguish it from control signaling.



From request to response

01
The mobile device generates user data or a signaling request.

02
The radio access network carries data over radio, controls resources.

03
The core network manages connectivity, authentication, mobility, and routing.

04
The destination service processes the request and returns a response.

Conceptual view Uplink and downlink are shown as separate lanes for clarity. Control signaling is bidirectional.

PROTOCOL REFERENCE

Protocol Layers

Eight protocols transport mobile access and control traffic reliably.

Endpoint & packet network layers

Applications use TCP or UDP to transport IP between endpoints.

Radio access processing

PDCP, RLC and MAC prepare radio frames and schedule transmission.

Control-plane protocols

RRC controls the radio link while NAS connects the device to core services.

Network layer

IP

Internet Protocol

Addresses and routes packets between networks.

View protocol details ›

Transport layer

TCP

Transmission Control Protocol

Provides reliable, ordered data delivery.

View protocol details ›

Transport layer

UDP

User Datagram Protocol

Provides connectionless data transport.

View protocol details ›

Radio layer

PDCP

Packet Data Convergence Protocol

Compresses headers and provides security.

View protocol details ›

Radio layer

RLC

Radio Link Control

Supports segmentation, reassembly, and radio link recovery.

View protocol details ›

Radio layer

MAC

Medium Access Control

Manages radio resource access and scheduling.

View protocol details ›

Radio control

RRC

Radio Resource Control

Controls radio connections and state transitions.

View protocol details ›

Core signaling

NAS

Non-Access Stratum

Carries signaling between the mobile device and core network.

View protocol details ›

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